

ELITE MATHEMATICS

Pearson Edexcel International A Level

Pure 1

Adaptive Classified

Chapter 12 · Differentiation

With Answers

Questions with full worked solutions

- 69 pages
- Past papers up to June 2026

Differentiation

Your classified route

26
questions

208
marks

5
families

10
variants

DR ESLAM AHMED

CLASSIFIED BY IDEA

Move from recognition to method, then use the working space to complete the solution.

Differentiation

Families and variants

F01 • Power-rule differentiation and algebraic preparation

V01 Rewrite then find the first derivative

V02 Second derivative and higher-derivative comparison

F02 • Tangents, normals and linked line geometry

V01 Tangent or normal at a given point

V02 Find the contact point from a gradient or line

V03 Linked normals, tangents and parallel-gradient points

F03 • Equations, inequalities and extrema involving derivatives

V01 Stationary-point equation

V02 Solve a prescribed first- or second-derivative value

V03 Gradient inequality or extremum of the derivative

F04 • Chord gradients and the derivative limit

V01 Chord gradient tending to the tangent gradient

F05 • Global curve synthesis with tangents

V01 Tangent re-intersects the original curve

Power-rule differentiation and algebraic preparation

WHAT THIS FAMILY ASKS

Rewrite radicals and quotients as powers, differentiate term by term and simplify exact results.

VARIANT ROUTE

V01 Rewrite then find the first derivative

V02 Second derivative and higher-derivative comparison

3 Complete Classified questions

17 total marks

1. A curve C has equation

$$y = 2 + 10x^{\frac{1}{2}} - 2x^{\frac{3}{2}} \quad x > 0$$

- (a) Find $\frac{dy}{dx}$ giving your answer in simplest form. (3)
- (b) Hence find the exact value of the gradient of the tangent to C at the point where $x = 2$ giving your answer in simplest form.

(Solutions relying on calculator technology are not acceptable.) (2)

(Total for Question 1 is 5 marks)

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Question 1

WMA11/01 January 2023

T12 Differentiation

F01 Power-rule differentiation
and algebraic preparationV01 Rewrite then find the first
derivative

1

Differentiate term by term

$$y = 2 + 10x^{1/2} - 2x^{3/2}, \text{ so } \frac{dy}{dx} = 5x^{-1/2} - 3x^{1/2}.$$

2

Write in simplest form

$$\frac{dy}{dx} = \frac{5}{\sqrt{x}} - 3\sqrt{x}.$$

3

Substitute $x = 2$

$$\text{At } x = 2, \text{ the gradient is } \frac{5}{\sqrt{2}} - 3\sqrt{2} = \frac{5}{\sqrt{2}} - \frac{6}{\sqrt{2}} = -\frac{1}{\sqrt{2}}.$$

4

Rationalise

$$-\frac{1}{\sqrt{2}} = -\frac{\sqrt{2}}{2}.$$

FINAL ANSWER

$$(a) \frac{dy}{dx} = \frac{5}{\sqrt{x}} - 3\sqrt{x}. \quad (b) -\frac{\sqrt{2}}{2}.$$

4.

In this question you must show all stages of your working.

(a) Write

$$y = \frac{5x^2 + \sqrt{x^3}}{\sqrt[3]{8x}}$$

in the form

$$y = Ax^p + Bx^q$$

where A , B , p and q are constants to be found.

(4)

(b) Hence find $\frac{dy}{dx}$ giving each coefficient in simplest form.

(3)

(Total for Question 4 is 7 marks)

Question 2

WMA11/01 May/June 2023

T12 Differentiation

F01 Power-rule differentiation
and algebraic preparationV01 Rewrite then find the first
derivative

1

Rewrite the denominator

$$\sqrt[3]{8x} = 2x^{1/3}.$$

2

Rewrite the numerator

$$\sqrt{x^3} = x^{3/2}.$$

3

Divide each term

$$y = \frac{5x^2}{2x^{1/3}} + \frac{x^{3/2}}{2x^{1/3}} = \frac{5}{2}x^{5/3} + \frac{1}{2}x^{7/6}.$$

4

Differentiate

$$\frac{dy}{dx} = \frac{5}{2} \cdot \frac{5}{3}x^{2/3} + \frac{1}{2} \cdot \frac{7}{6}x^{1/6}.$$

5

Simplify coefficients

$$\frac{dy}{dx} = \frac{25}{6}x^{2/3} + \frac{7}{12}x^{1/6}.$$

FINAL ANSWER

$$(a) y = \frac{5}{2}x^{5/3} + \frac{1}{2}x^{7/6}. \quad (b) \frac{dy}{dx} = \frac{25}{6}x^{2/3} + \frac{7}{12}x^{1/6}.$$

1. Given that

$$y = 5x^3 + \frac{3}{x^2} - 7x \quad x > 0$$

find, in simplest form,

(a) $\frac{dy}{dx}$ (3)

(b) $\frac{d^2y}{dx^2}$ (2)

(Total for Question 1 is 5 marks)

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Question 3

WMA11/01 October 2023

T12 Differentiation

F01 Power-rule differentiation
and algebraic preparationV02 Second derivative and
higher-derivative comparison

1

Rewrite the powers

$$y = 5x^3 + 3x^{-2} - 7x.$$

2

Differentiate once

$$\frac{dy}{dx} = 15x^2 - 6x^{-3} - 7 = 15x^2 - \frac{6}{x^3} - 7.$$

3

Differentiate again

$$\frac{d^2y}{dx^2} = 30x + 18x^{-4} = 30x + \frac{18}{x^4}.$$

FINAL ANSWER

$$\frac{dy}{dx} = 15x^2 - \frac{6}{x^3} - 7, \text{ and } \frac{d^2y}{dx^2} = 30x + \frac{18}{x^4}.$$

Tangents, normals and linked line geometry

WHAT THIS FAMILY ASKS

Use derivative gradients with point-gradient form, reciprocal gradients and linked tangent or normal conditions.

VARIANT ROUTE

V01 Tangent or normal at a given point

V02 Find the contact point from a gradient or line

V03 Linked normals, tangents and parallel-gradient points

11 Complete Classified questions

96 total marks

5. A curve has equation

$$y = \frac{x^3}{6} + 4\sqrt{x} - 15 \quad x \geq 0$$

(a) Find $\frac{dy}{dx}$, giving the answer in simplest form.

(3)

The point $P\left(4, \frac{11}{3}\right)$ lies on the curve.

(b) Find the equation of the normal to the curve at P . Write your answer in the form $ax + by + c = 0$, where a , b and c are integers to be found.

(4)

(Total 7 marks)

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Question 4

WMA11/01 October 2019

T12 Differentiation

F02 Tangents, normals and linked line geometry

V01 Tangent or normal at a given point

1

Rewrite before differentiating

$$y = \frac{x^3}{6} + 4x^{1/2} - 15.$$

2

Differentiate term by term

$$\frac{dy}{dx} = \frac{3x^2}{6} + 4 \cdot \frac{1}{2}x^{-1/2} = \frac{x^2}{2} + \frac{2}{\sqrt{x}}.$$

3

Find tangent and normal gradients

At $x = 4$, $\frac{dy}{dx} = \frac{16}{2} + \frac{2}{2} = 9$. The normal gradient is therefore $-\frac{1}{9}$.

4

Build the normal equation

Using $P(4, \frac{11}{3})$: $y - \frac{11}{3} = -\frac{1}{9}(x - 4)$. Multiplying out gives $x + 9y - 37 = 0$.

FINAL ANSWER

$$\frac{dy}{dx} = \frac{x^2}{2} + \frac{2}{\sqrt{x}}, \text{ and the normal is } x + 9y - 37 = 0.$$

1. A curve has equation

$$y = 2x^3 - 5x^2 - \frac{3}{2x} + 7 \quad x > 0$$

(a) Find, in simplest form, $\frac{dy}{dx}$ (3)

The point P lies on the curve and has x coordinate $\frac{1}{2}$

(b) Find an equation of the normal to the curve at P , writing your answer in the form $ax + by + c = 0$, where a , b and c are integers to be found. (5)

(Total 8 marks)

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Question 5

WMA11/01 January 2021

T12 Differentiation

F02 Tangents, normals and linked line geometry

V01 Tangent or normal at a given point

1

Rewrite the negative power term

$$y = 2x^3 - 5x^2 - \frac{3}{2x} + 7 = 2x^3 - 5x^2 - \frac{3}{2}x^{-1} + 7.$$

2

Differentiate term by term

$$\frac{dy}{dx} = 6x^2 - 10x + \frac{3}{2}x^{-2} = 6x^2 - 10x + \frac{3}{2x^2}.$$

3

Find the point P

$$\text{At } x = \frac{1}{2}, y = 2\left(\frac{1}{8}\right) - 5\left(\frac{1}{4}\right) - \frac{3}{1} + 7 = 3, \text{ so } P = \left(\frac{1}{2}, 3\right).$$

4

Find the normal gradient

$$\text{At } x = \frac{1}{2}, \frac{dy}{dx} = 6\left(\frac{1}{4}\right) - 10\left(\frac{1}{2}\right) + \frac{3}{2(1/4)} = \frac{5}{2}. \text{ The normal gradient is therefore } -\frac{2}{5}.$$

5

Write the normal in integer form

$$y - 3 = -\frac{2}{5}\left(x - \frac{1}{2}\right). \text{ Multiplying by 5 gives } 5y - 15 = -2x + 1, \text{ hence } 2x + 5y - 16 = 0.$$

FINAL ANSWER

$$(a) \frac{dy}{dx} = 6x^2 - 10x + \frac{3}{2x^2}. \quad (b) 2x + 5y - 16 = 0.$$

1. The curve C has equation

$$y = \frac{x^2}{3} + \frac{4}{\sqrt{x}} + \frac{8}{3x} - 5 \quad x > 0$$

- (a) Find $\frac{dy}{dx}$, giving your answer in simplest form.

(4)

The point $P(4,3)$ lies on C .

- (b) Find the equation of the normal to C at the point P . Write your answer in the form $ax + by + c = 0$, where a , b and c are integers to be found.

(4)

(Total 8 marks)

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Question 6

WMA11/01 May/June 2021

T12 Differentiation

F02 Tangents, normals and linked line geometry

V01 Tangent or normal at a given point

1

Rewrite using powers

$$y = \frac{x^2}{3} + 4x^{-1/2} + \frac{8}{3}x^{-1} - 5.$$

2

Differentiate term by term

$$\frac{dy}{dx} = \frac{2x}{3} - 2x^{-3/2} - \frac{8}{3}x^{-2} = \frac{2x}{3} - \frac{2}{x\sqrt{x}} - \frac{8}{3x^2}.$$

3

Find the tangent gradient at P

$$\text{At } x = 4, \frac{dy}{dx} = \frac{8}{3} - \frac{1}{4} - \frac{1}{6} = \frac{9}{4}.$$

4

Find the normal gradient

The normal gradient is the negative reciprocal of $\frac{9}{4}$, so $m_n = -\frac{4}{9}$.

5

Write the normal equation

Using $P(4, 3)$, $y - 3 = -\frac{4}{9}(x - 4)$. Multiplying by 9 gives $9y - 27 = -4x + 16$, hence $4x + 9y - 43 = 0$.

FINAL ANSWER

$$(a) \frac{dy}{dx} = \frac{2x}{3} - \frac{2}{x\sqrt{x}} - \frac{8}{3x^2}. \quad (b) 4x + 9y - 43 = 0.$$

1. The curve C has equation

$$y = \frac{x^3}{4} - x^2 + \frac{17}{x} \quad x > 0$$

(a) Find $\frac{dy}{dx}$, giving your answer in simplest form.

(3)

The point $R\left(2, \frac{13}{2}\right)$ lies on C .

(b) Find the equation of the tangent to C at the point R . Write your answer in the form $ax + by + c = 0$, where a , b and c are integers to be found.

(3)

(Total for Question 1 is 6 marks)

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Question 7

WMA11/01 October 2022

T12 Differentiation

F02 Tangents, normals and linked line geometry

V01 Tangent or normal at a given point

1

Rewrite the curve

$$y = \frac{x^3}{4} - x^2 + \frac{17}{x} = \frac{1}{4}x^3 - x^2 + 17x^{-1}.$$

2

Differentiate

$$\frac{dy}{dx} = \frac{3}{4}x^2 - 2x - 17x^{-2} = \frac{3}{4}x^2 - 2x - \frac{17}{x^2}.$$

3

Find the gradient at R

$$\text{At } x = 2, \text{ the tangent gradient is } \frac{3}{4}(4) - 2(2) - \frac{17}{4} = 3 - 4 - \frac{17}{4} = -\frac{21}{4}.$$

4

Use the point R

$$\text{The point is } R\left(2, \frac{13}{2}\right), \text{ so } y - \frac{13}{2} = -\frac{21}{4}(x - 2).$$

5

Write in integer form

$$\text{Multiplying by 4 gives } 4y - 26 = -21x + 42, \text{ hence } 21x + 4y - 68 = 0.$$

FINAL ANSWER

$$(a) \frac{dy}{dx} = \frac{3}{4}x^2 - 2x - \frac{17}{x^2}. \quad (b) 21x + 4y - 68 = 0.$$

3. The curve C has equation

$$y = \frac{5x^3 - 8}{2x^2} \quad x > 0$$

- (a) Find $\frac{dy}{dx}$ writing your answer in simplest form. (4)

The point $P(2, 4)$ lies on C .

- (b) Find an equation for the tangent to C at P writing your answer in the form $ax + by + c = 0$, where a , b and c are integers. (3)

(Total for Question 3 is 7 marks)

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Question 8

WMA11/01 January 2024

T12 Differentiation

F02 Tangents, normals and linked line geometry

V01 Tangent or normal at a given point

1

Rewrite y

$$y = \frac{5x^3 - 8}{2x^2} = \frac{5}{2}x - 4x^{-2}.$$

2

Differentiate

$$\frac{dy}{dx} = \frac{5}{2} + 8x^{-3} = \frac{5}{2} + \frac{8}{x^3}.$$

3

Find the gradient at P

$$\text{At } x = 2, \frac{dy}{dx} = \frac{5}{2} + \frac{8}{8} = \frac{7}{2}.$$

4

Use point-gradient form

$$\text{The tangent through } P(2, 4) \text{ is } y - 4 = \frac{7}{2}(x - 2).$$

5

Rearrange

$$2y - 8 = 7x - 14, \text{ so } 7x - 2y - 6 = 0.$$

FINAL ANSWER

$$(a) \frac{dy}{dx} = \frac{5}{2} + \frac{8}{x^3}. \quad (b) 7x - 2y - 6 = 0.$$

1. The curve C has equation

$$y = 6x^2 + 3\sqrt{x} + \frac{5}{8} \quad x > 0$$

(a) Find $\frac{dy}{dx}$ writing the answer in simplest form.

(3)

The point $P\left(\frac{1}{4}, \frac{5}{2}\right)$ lies on C .

(b) Find the equation of the tangent to C at P , writing your answer in the form $y = mx + c$ where m and c are integers.

(3)

(Total for Question 1 is 6 marks)

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Question 9

WMA11/01 October 2025

T12 Differentiation

F02 Tangents, normals and linked line geometry

V01 Tangent or normal at a given point

1

Rewrite the curve

$$y = 6x^2 + 3x^{1/2} + \frac{5}{8}.$$

2

Differentiate

$$\frac{dy}{dx} = 12x + \frac{3}{2}x^{-1/2} = 12x + \frac{3}{2\sqrt{x}}.$$

3

Find the tangent gradient

$$\text{At } x = \frac{1}{4}, \frac{dy}{dx} = 12\left(\frac{1}{4}\right) + \frac{3}{2}\left(\frac{1}{4}\right)^{-1/2} = 3 + 3 = 6.$$

4

Use point P

$$\text{The tangent passes through } P\left(\frac{1}{4}, \frac{5}{2}\right), \text{ so } y - \frac{5}{2} = 6\left(x - \frac{1}{4}\right).$$

5

Simplify

$$y - \frac{5}{2} = 6x - \frac{3}{2}, \text{ hence } y = 6x + 1.$$

FINAL ANSWER

$$(a) \frac{dy}{dx} = 12x + \frac{3}{2\sqrt{x}}. \quad (b) y = 6x + 1.$$

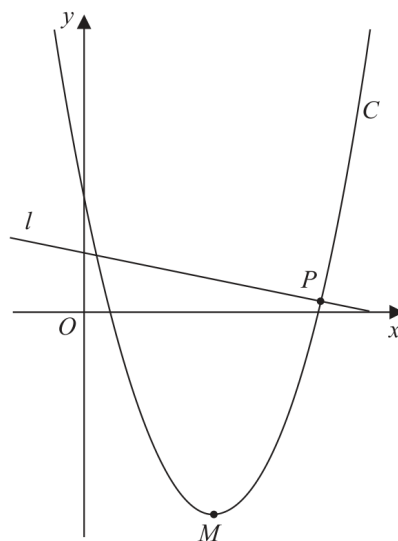


Figure 3

Figure 3 shows a sketch of the curve C with equation

$$y = \frac{1}{2}x^2 - 10x + 22$$

- (a) Write $\frac{1}{2}x^2 - 10x + 22$ in the form

$$a(x + b)^2 + c$$

where a , b and c are constants to be found.

(3)

The point M is the minimum turning point of C , as shown in Figure 3.

- (b) Deduce the coordinates of M

(2)

The line l is the normal to C at the point P , as shown in Figure 3.

Given that l has equation $y = k - \frac{1}{8}x$, where k is a constant,

- (c) (i) find the coordinates of P

- (ii) find the value of k

(6)

Question 9 continues on the next page

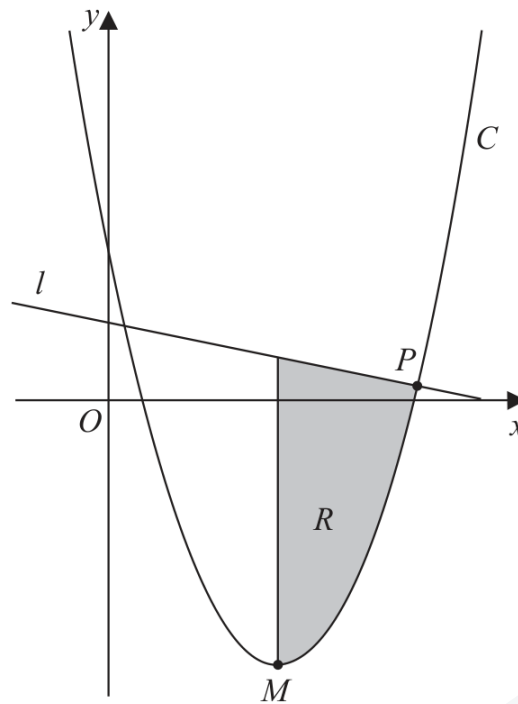


Figure 4

Figure 4 is a copy of Figure 3. The finite region R , shown shaded in Figure 4, is bounded by l , C and the line through M parallel to the y -axis.

(d) Identify the inequalities that define R .

(3)

(Total for Question 9 is 14 marks)

Question 10

WMA11/01 October 2022

T12 Differentiation

F02 Tangents, normals and linked line geometry

V02 Find the contact point from a gradient or line

1

Complete the square

$$\frac{1}{2}x^2 - 10x + 22 = \frac{1}{2}(x^2 - 20x) + 22 = \frac{1}{2}((x - 10)^2 - 100) + 22.$$

2

Simplify the completed square form

$$\frac{1}{2}((x - 10)^2 - 100) + 22 = \frac{1}{2}(x - 10)^2 - 28.$$

3

Read the minimum pointThe minimum point is $M(10, -28)$.

4

Use the normal gradientThe line l has gradient $-\frac{1}{8}$, so the tangent gradient at P is 8.

5

Differentiate the curve

$$\frac{dy}{dx} = x - 10. \text{ At } P, x - 10 = 8, \text{ so } x = 18.$$

6

Find P

$$\text{When } x = 18, y = \frac{1}{2}(18)^2 - 10(18) + 22 = 4, \text{ so } P = (18, 4).$$

7

Find k

$$\text{Since } P \text{ lies on } y = k - \frac{1}{8}x, 4 = k - \frac{1}{8}(18) = k - \frac{9}{4}, \text{ so } k = \frac{25}{4}.$$

8

Write the upper boundaryThe line is $y = \frac{25}{4} - \frac{x}{8}$.

9

Write the lower boundaryThe curve is $y = \frac{1}{2}x^2 - 10x + 22$.

10

Identify the regionThe region lies to the right of the vertical line through M and between the curve and the normal line: $10 \leq x \leq 18$ and $\frac{1}{2}x^2 - 10x + 22 \leq y \leq \frac{25}{4} - \frac{x}{8}$.**FINAL ANSWER**

(a) $\frac{1}{2}(x - 10)^2 - 28$. (b) $M(10, -28)$. (c) $P(18, 4)$, $k = \frac{25}{4}$. (d) $10 \leq x \leq 18$, $\frac{1}{2}x^2 - 10x + 22 \leq y \leq \frac{25}{4} - \frac{x}{8}$.

5.

**In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.**

The curve C has equation

$$y = 4x^3 + \frac{2}{x} + 9 \quad x > 0$$

(a) Find $\frac{dy}{dx}$, giving your answer in simplest form.

(2)

Given that

- the point P lies on C
- the line with equation $y = k - 5x$, where k is a constant, is the tangent to C at P

(b) show that the x coordinate of P satisfies the equation

$$12x^4 + 5x^2 - 2 = 0$$

(2)

(c) Hence find the value of k .

(4)

(Total for Question 5 is 8 marks)

Question 11

WMA11/01 January 2025

T12 Differentiation

F02 Tangents, normals and linked line geometry

V02 Find the contact point from a gradient or line

1

Differentiate

$$y = 4x^3 + 2x^{-1} + 9, \text{ so } \frac{dy}{dx} = 12x^2 - 2x^{-2} = 12x^2 - \frac{2}{x^2}.$$

2

Use the tangent gradient

$$\text{The tangent line } y = k - 5x \text{ has gradient } -5, \text{ so at } P, 12x^2 - \frac{2}{x^2} = -5.$$

3

Clear the denominator

$$\text{Multiplying by } x^2 \text{ gives } 12x^4 - 2 = -5x^2, \text{ hence } 12x^4 + 5x^2 - 2 = 0.$$

4

Solve for x squared

$$\text{Let } u = x^2. \text{ Then } 12u^2 + 5u - 2 = (3u + 2)(4u - 1) = 0.$$

5

Use x greater than zero

$$x^2 = \frac{1}{4}, \text{ so } x = \frac{1}{2}.$$

6

Find the y-coordinate of P

$$y = 4\left(\frac{1}{2}\right)^3 + \frac{2}{1/2} + 9 = \frac{1}{2} + 4 + 9 = \frac{27}{2}.$$

7

Find k

Since P lies on $y = k - 5x$, $\frac{27}{2} = k - \frac{5}{2}$, so $k = 16$.

FINAL ANSWER

(a) $\frac{dy}{dx} = 12x^2 - \frac{2}{x^2}$. (c) $k = 16$.

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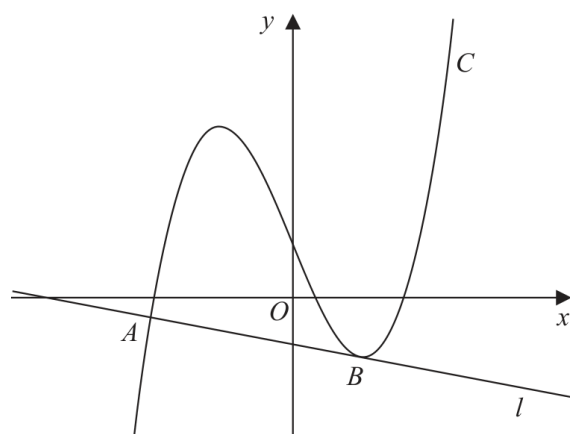


Figure 5

Figure 5 shows a sketch of the curve C with equation

$$y = \frac{2}{7}x^3 + \frac{1}{7}x^2 - \frac{5}{2}x + k$$

where k is a constant.

- (a) Find $\frac{dy}{dx}$ (2)

The line l , shown in Figure 5, is the normal to C at the point A with x coordinate $-\frac{7}{2}$

Given that l is also a tangent to C at the point B ,

- (b) show that the x coordinate of the point B is a solution of the equation

$$12x^2 + 4x - 33 = 0 \quad (4)$$

- (c) Hence find the x coordinate of B , justifying your answer. (2)

Given that the y intercept of l is -1

- (d) find the value of k . (4)

(Total 12 marks)

Question 12

WMA11/01 May/June 2022

T12 Differentiation

F02 Tangents, normals and linked line geometry

V03 Linked normals, tangents and parallel-gradient points

1

Differentiate the curve

For $y = \frac{2}{7}x^3 + \frac{1}{7}x^2 - \frac{5}{2}x + k$, $\frac{dy}{dx} = \frac{6}{7}x^2 + \frac{2}{7}x - \frac{5}{2}$.

2

Find the tangent gradient at A

At $x = -\frac{7}{2}$, $\frac{dy}{dx} = \frac{6}{7} \left(\frac{49}{4} \right) + \frac{2}{7} \left(-\frac{7}{2} \right) - \frac{5}{2} = \frac{21}{2} - 1 - \frac{5}{2} = 7$.

3

Find the gradient of l

Since l is the normal at A , its gradient is the negative reciprocal of 7, so $m_l = -\frac{1}{7}$.

4

Use that l is tangent at B

At B , the tangent gradient is also $-\frac{1}{7}$, so $\frac{6}{7}x^2 + \frac{2}{7}x - \frac{5}{2} = -\frac{1}{7}$.

5

Reach the given quadratic

Multiplying by 14 gives $12x^2 + 4x - 35 = -2$, hence $12x^2 + 4x - 33 = 0$.

6

Solve for the x-coordinate of B

$12x^2 + 4x - 33 = 0$ gives $x = \frac{-4 \pm 40}{24}$, so $x = \frac{3}{2}$ or $x = -\frac{11}{6}$. From the diagram, B is the point to the right of the y-axis, so $x_B = \frac{3}{2}$.

7

Use the y-intercept of l

Since the y-intercept is -1 and the gradient is $-\frac{1}{7}$, the line is $y = -\frac{1}{7}x - 1$.

8

Find the y-coordinate of A

At $x = -\frac{7}{2}$ on the line, $y = -\frac{1}{7}\left(-\frac{7}{2}\right) - 1 = \frac{1}{2} - 1 = -\frac{1}{2}$.

9

Substitute A into the curve

$$-\frac{1}{2} = \frac{2}{7}\left(-\frac{7}{2}\right)^3 + \frac{1}{7}\left(-\frac{7}{2}\right)^2 - \frac{5}{2}\left(-\frac{7}{2}\right) + k = -\frac{7}{4} + k.$$

10

Find k

Therefore $k = \frac{5}{4}$.

FINAL ANSWER

$$(a) \frac{dy}{dx} = \frac{6}{7}x^2 + \frac{2}{7}x - \frac{5}{2}. \quad (c) x_B = \frac{3}{2}. \quad (d) k = \frac{5}{4}.$$

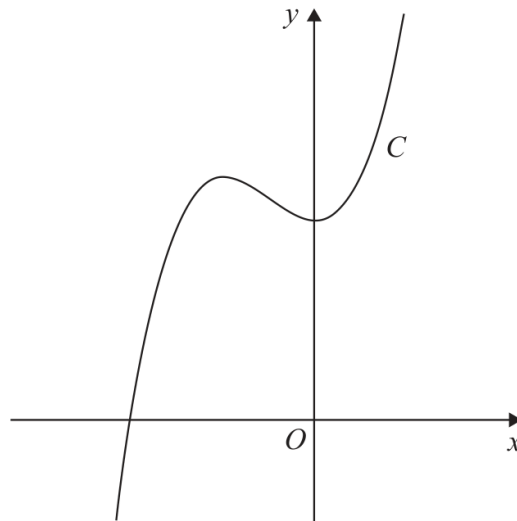


Figure 4

Figure 4 shows a sketch of the curve C with equation $y = f(x)$, where

$$f(x) = (x + 5)(3x^2 - 4x + 20)$$

(a) Deduce the range of values of x for which $f(x) \geq 0$ (1)

(b) Find $f'(x)$ giving your answer in simplest form. (3)

The point $R(-4, 84)$ lies on C .

Given that the tangent to C at the point P is parallel to the tangent to C at the point R

(c) find the x coordinate of P . (4)

(d) Find the point to which R is transformed when the curve with equation $y = f(x)$ is transformed to the curve with equation,

(i) $y = f(x - 3)$

(ii) $y = 4f(x)$ (2)

(Total for Question 9 is 10 marks)

Question 13

WMA11/01 October 2024

T12 Differentiation

F02 Tangents, normals and linked line geometry

V03 Linked normals, tangents and parallel-gradient points

1

Use the positive quadratic factor

$3x^2 - 4x + 20$ has discriminant $(-4)^2 - 4(3)(20) < 0$ and positive leading coefficient, so it is always positive.

2

Deduce the sign of $f(x)$

The sign of $f(x) = (x + 5)(3x^2 - 4x + 20)$ is therefore the sign of $x + 5$. Hence $f(x) \geq 0$ for $x \geq -5$.

3

Expand $f(x)$

$$f(x) = 3x^3 + 11x^2 + 100.$$

4

Differentiate

$$f'(x) = 9x^2 + 22x.$$

5

Find the gradient at R

At $R(-4, 84)$, $f'(-4) = 9(16) + 22(-4) = 56$.

6

Use parallel tangents

For the tangent at P to be parallel, $f'(x) = 56$.

7

Solve for x

$9x^2 + 22x - 56 = 0$, so $x = \frac{-22 \pm 50}{18}$. The roots are $x = -4$ and $x = \frac{14}{9}$.

8

Choose the point P

$x = -4$ is the point R, so $x_P = \frac{14}{9}$.

9

Translate point R

$y = f(x - 3)$ translates the curve 3 units to the right, so $(-4, 84)$ maps to $(-1, 84)$.

10

Transform R under y equals 4f(x)

$y = 4f(x)$ stretches vertically by scale factor 4, so $(-4, 84)$ maps to $(-4, 336)$.

FINAL ANSWER

(a) $x \geq -5$. (b) $f'(x) = 9x^2 + 22x$. (c) $x_P = \frac{14}{9}$. (d) (i) $(-1, 84)$.
(d) (ii) $(-4, 336)$.

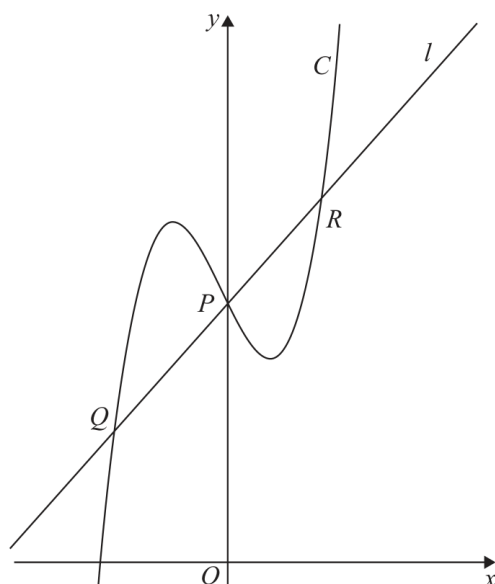


Figure 4

In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

Figure 4 shows a sketch of the curve C with equation

$$y = 2x^3 + \frac{1}{2}x^2 - 2x + 5$$

The line l is the normal to C at the point P where $x = 0$

The line l also intersects C at points Q and R as shown in Figure 4.

(a) Find, using algebra, the x coordinate of point Q .

(6)

The point T lies on C .

Given that

- the tangent to C at T is parallel to l
- the x coordinate of T is positive

(b) find, using algebra, the exact x coordinate of T .

(4)

(Total for Question 10 is 10 marks)

Question 14

WMA11/01 October 2025

T12 Differentiation

F02 Tangents, normals and linked line geometry

V03 Linked normals, tangents and parallel-gradient points

1

Differentiate C

$$y = 2x^3 + \frac{1}{2}x^2 - 2x + 5, \text{ so } \frac{dy}{dx} = 6x^2 + x - 2.$$

2

Find the tangent gradient at PAt $P, x = 0$, so the tangent gradient is -2 .

3

Equation of the normalThe normal gradient is $\frac{1}{2}$ and $P = (0, 5)$, so $l : y = \frac{1}{2}x + 5$.

4

Find intersections with the normal

$$\text{Set } 2x^3 + \frac{1}{2}x^2 - 2x + 5 = \frac{1}{2}x + 5.$$

5

Form and factor the cubic

$$2x^3 + \frac{1}{2}x^2 - \frac{5}{2}x = 0. \text{ Multiplying by 2 gives } 4x^3 + x^2 - 5x = 0 = x(4x^2 + x - 5).$$

6

Solve for x

$$x(4x + 5)(x - 1) = 0, \text{ so the intersections have x-coordinates } 0, -\frac{5}{4} \text{ and } 1.$$

7

Select point Q

$$\text{The point } Q \text{ is the left-hand intersection, so } x_Q = -\frac{5}{4}.$$

8

Parallel tangent condition

The tangent at T is parallel to l , so its gradient is $\frac{1}{2}$.

9

Set derivative equal to one half

$$6x^2 + x - 2 = \frac{1}{2}, \text{ so } 12x^2 + 2x - 5 = 0.$$

10

Use the positive solution

$$x = \frac{-2 \pm \sqrt{4 + 240}}{24} = \frac{-1 \pm \sqrt{61}}{12}. \text{ Since } x_T \text{ is positive, } x_T = \frac{-1 + \sqrt{61}}{12}.$$

FINAL ANSWER

$$(a) x_Q = -\frac{5}{4}. \quad (b) x_T = \frac{-1 + \sqrt{61}}{12}.$$

Equations, inequalities and extrema involving derivatives

WHAT THIS FAMILY ASKS

Solve conditions imposed on f' or f'' and interpret domain, sign or minimum-gradient information.

VARIANT ROUTE

V01 Stationary-point equation

V02 Solve a prescribed first- or second-derivative value

V03 Gradient inequality or extremum of the derivative

9 Complete Classified questions

73 total marks

9.

In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

A curve has equation

$$y = \frac{4x^2 + 9}{2\sqrt{x}} \quad x > 0$$

Find the x coordinate of the point on the curve at which $\frac{dy}{dx} = 0$

(6)

(Total 6 marks)

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Question 15

WMA11/01 January 2020

T12 Differentiation

F03 Equations, inequalities and extrema involving derivatives

V01 Stationary-point equation

1

Rewrite the function

$$y = \frac{4x^2 + 9}{2\sqrt{x}} = 2x^{3/2} + \frac{9}{2}x^{-1/2}.$$

2

Differentiate

$$\frac{dy}{dx} = 3x^{1/2} - \frac{9}{4}x^{-3/2}.$$

3

Set the derivative equal to zero

$$3x^{1/2} - \frac{9}{4}x^{-3/2} = 0, \text{ so } 3\sqrt{x} = \frac{9}{4x^{3/2}}.$$

4

Solve for x

Multiply by $4x^{3/2}$: $12x^2 = 9$. **Hence** $x^2 = \frac{3}{4}$. **Since** $x > 0$, $x = \frac{\sqrt{3}}{2}$.

FINAL ANSWER

$$x = \frac{\sqrt{3}}{2}.$$

3.

In this question you must show all stages of your working.

$$f(x) = \frac{(x+5)^2}{\sqrt{x}} \quad x > 0$$

(a) Find $\int f(x) \, dx$ (4)

(b) (i) Show that when $f'(x) = 0$

$$3x^2 + 10x - 25 = 0$$

(ii) Hence state the value of x for which

$$f'(x) = 0$$
 (5)

(Total for Question 3 is 9 marks)

Question 16

WMA11/01 October 2024

T12 Differentiation

F03 Equations, inequalities and extrema involving derivatives

V01 Stationary-point equation

1

Rewrite $f(x)$

$$f(x) = \frac{(x+5)^2}{\sqrt{x}} = \frac{x^2 + 10x + 25}{x^{1/2}} = x^{3/2} + 10x^{1/2} + 25x^{-1/2}.$$

2

Integrate

$$\int f(x) dx = \frac{2}{5}x^{5/2} + \frac{20}{3}x^{3/2} + 50x^{1/2} + c.$$

3

Differentiate $f(x)$

$$f'(x) = \frac{3}{2}x^{1/2} + 5x^{-1/2} - \frac{25}{2}x^{-3/2}.$$

4

Set f' prime equal to zero

$$\frac{3}{2}x^{1/2} + 5x^{-1/2} - \frac{25}{2}x^{-3/2} = 0.$$

5

Clear powers of x Multiplying by $2x^{3/2}$ gives $3x^2 + 10x - 25 = 0$.

6

Use x greater than zero

$$3x^2 + 10x - 25 = (3x - 5)(x + 5), \text{ so } x = \frac{5}{3} \text{ or } x = -5. \text{ Since } x > 0, x = \frac{5}{3}.$$

FINAL ANSWER

$$(a) \frac{2}{5}x^{5/2} + \frac{20}{3}x^{3/2} + 50x^{1/2} + c. \quad (b) x = \frac{5}{3}.$$

7.

In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

$$f(x) = 2x - 3\sqrt{x} - 5 \quad x > 0$$

(a) Solve the equation

$$f(x) = 9 \quad (4)$$

(b) Solve the equation

$$f''(x) = 6 \quad (5)$$

(Total 9 marks)

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Question 17

WMA11/01 January 2021

T12 Differentiation

F03 Equations, inequalities and extrema involving derivatives

V02 Solve a prescribed first- or second-derivative value

1

Solve $f(x) = 9$ without calculator methods $2x - 3\sqrt{x} - 5 = 9$, so $2x - 3\sqrt{x} - 14 = 0$.

2

Use $u = \sqrt{x}$ Let $u = \sqrt{x}$, where $u > 0$. Then $x = u^2$, so $2u^2 - 3u - 14 = 0$.

3

Factorise and reject the invalid root $2u^2 - 3u - 14 = (2u - 7)(u + 2)$. Since $u > 0$, $u = \frac{7}{2}$, so $x = u^2 = \frac{49}{4}$.

4

Differentiate twice $f(x) = 2x - 3x^{1/2} - 5$, so $f'(x) = 2 - \frac{3}{2}x^{-1/2}$ and $f''(x) = \frac{3}{4}x^{-3/2}$.

5

Solve $f''(x) = 6$ $\frac{3}{4}x^{-3/2} = 6$, so $x^{-3/2} = 8$. Therefore $x^{3/2} = \frac{1}{8}$.

6

Find x Since $x^{3/2} = (\sqrt{x})^3$, we get $\sqrt{x} = \frac{1}{2}$, hence $x = \frac{1}{4}$.

FINAL ANSWER

$$(a) x = \frac{49}{4}. \quad (b) x = \frac{1}{4}.$$

2. **In this question you must show all stages of your working.**

Solutions relying on calculator technology are not acceptable.

A curve has equation

$$y = 3x^5 + 4x^3 - x + 5$$

The points P and Q lie on the curve.

The gradient of the curve at both point P and point Q is 2

Find the x coordinates of P and Q .

(5)

(Total 5 marks)

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Question 18

WMA11/01 October 2021

T12 Differentiation

F03 Equations, inequalities and extrema involving derivatives

V02 Solve a prescribed first- or second-derivative value

1

Differentiate the curveFor $y = 3x^5 + 4x^3 - x + 5$, $\frac{dy}{dx} = 15x^4 + 12x^2 - 1$.

2

Set the gradient equal to 2 $15x^4 + 12x^2 - 1 = 2$, so $15x^4 + 12x^2 - 3 = 0$.

3

Reduce to a quadratic in x squaredDividing by 3 gives $5x^4 + 4x^2 - 1 = 0$. Let $u = x^2$, giving $5u^2 + 4u - 1 = 0$.

4

Solve and reject the impossible value $5u^2 + 4u - 1 = (5u - 1)(u + 1) = 0$. Since $u = x^2 \geq 0$, $u = \frac{1}{5}$.

5

Find x $x^2 = \frac{1}{5}$, so $x = \pm \frac{1}{\sqrt{5}} = \pm \frac{\sqrt{5}}{5}$.**FINAL ANSWER**

$$x = \pm \frac{\sqrt{5}}{5}$$

7. The curve C has equation $y = f(x)$ where

$$f(x) = 2x^3 - kx^2 + 14x + 24$$

and k is a constant.

(a) Find, in simplest form,

(i) $f'(x)$

(ii) $f''(x)$

(3)

The curve with equation $y = f'(x)$ intersects the curve with equation $y = f''(x)$ at the points A and B .

Given that the x coordinate of A is 5

(b) find the value of k .

(2)

(c) Hence find the coordinates of B .

(3)

(Total for Question 7 is 8 marks)

Question 19

WMA11/01 May/June 2024

T12 Differentiation

F03 Equations, inequalities and extrema involving derivatives

V02 Solve a prescribed first- or second-derivative value

1

Differentiate once

$$f'(x) = 6x^2 - 2kx + 14.$$

2

Differentiate again

$$f''(x) = 12x - 2k.$$

3

Use the intersection at A

At an intersection of $y = f'(x)$ and $y = f''(x)$, $f'(x) = f''(x)$. Given $x_A = 5$, substitute $x = 5$.

4

Find k

$$6(5)^2 - 2k(5) + 14 = 12(5) - 2k, \text{ so } 164 - 10k = 60 - 2k \text{ and } k = 13.$$

5

Find the other x-coordinate

With $k = 13$, $6x^2 - 26x + 14 = 12x - 26$, so $6x^2 - 38x + 40 = 0$. Hence $(3x - 4)(x - 5) = 0$.

6

Use the point B

The other point has $x = \frac{4}{3}$. Then $y = f''(\frac{4}{3}) = 12(\frac{4}{3}) - 26 = -10$.

FINAL ANSWER

$$(a)(i) f'(x) = 6x^2 - 2kx + 14. \quad (a)(ii) f''(x) = 12x - 2k. \quad (b) k = 13.$$

$$(c) B = (\frac{4}{3}, -10).$$

2. The curve C has equation

$$y = 2x^{\frac{5}{2}} - 4x + 3$$

(a) Find $\frac{dy}{dx}$ writing your answer in simplest form.

(2)

The point P lies on C .

Given that

- the x coordinate of P is 2^k where k is a constant
- the gradient of C at the point P is 16

(b) find the value of k .

(3)

(Total for Question 2 is 5 marks)

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Question 20

WMA11/01 May/June 2025

T12 Differentiation

F03 Equations, inequalities and extrema involving derivatives

V02 Solve a prescribed first- or second-derivative value

1

Differentiate

$$y = 2x^{5/2} - 4x + 3, \text{ so } \frac{dy}{dx} = 2 \cdot \frac{5}{2}x^{3/2} - 4 = 5x^{3/2} - 4.$$

2

Use the gradient at PAt P, the gradient is 16, so $5x^{3/2} - 4 = 16$.

3

Find x

$$5x^{3/2} = 20, \text{ hence } x^{3/2} = 4.$$

4

Use x equals 2 to the kSince the x-coordinate of P is 2^k , we have $(2^k)^{3/2} = 4 = 2^2$.

5

Solve for k

$$2^{3k/2} = 2^2, \text{ so } \frac{3k}{2} = 2 \text{ and } k = \frac{4}{3}.$$

FINAL ANSWER

$$(a) \frac{dy}{dx} = 5x^{3/2} - 4. \quad (b) k = \frac{4}{3}.$$

4.

**In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.**

$$f(x) = x^3 - 2x^2\sqrt{x} + 14 \quad x > 0$$

(a) Solve the equation

$$f'(x) = 0 \quad (5)$$

(b) (i) Show that the equation

$$f''(x) = 66$$

can be written as

$$4x - 5x^{\frac{1}{2}} - 44 = 0$$

(ii) Hence, using algebra, show that $x = 16$ is the only solution of the equation

$$f''(x) = 66 \quad (5)$$

(Total for Question 4 is 10 marks)

Question 21

WMA11/01 May/June 2026

T12 Differentiation

F03 Equations, inequalities and extrema involving derivatives

V02 Solve a prescribed first- or second-derivative value

1

(a) Rewrite the radical as a power

$$f(x) = x^3 - 2x^{5/2} + 14.$$

2

(a) Differentiate and solve

$f'(x) = 3x^2 - 5x^{3/2} = x^{3/2}(3x^{1/2} - 5)$. Since $x > 0$, the factor $x^{3/2}$ is non-zero. Therefore $3\sqrt{x} - 5 = 0$, so $\sqrt{x} = \frac{5}{3}$ and $x = \frac{25}{9}$.

3

(b)(i) Differentiate again

$$f''(x) = 6x - \frac{15}{2}x^{1/2}.$$

4

(b)(i) Rearrange the given equation

$6x - \frac{15}{2}x^{1/2} = 66$. Multiplying by 2 and dividing by 3 gives $4x - 5x^{1/2} - 44 = 0$, as required.

5

(b)(ii) Factorise as a quadratic in the square root

Let $u = \sqrt{x}$, where $u \geq 0$. Then $4u^2 - 5u - 44 = 0$, and $(u - 4)(4u + 11) = 0$. Hence $u = 4$ or $u = -\frac{11}{4}$.

6

(b)(ii) Apply the square-root restriction

Because $u = \sqrt{x} \geq 0$, the value $-\frac{11}{4}$ is impossible. Thus $\sqrt{x} = 4$ and $x = 16$, which is the only solution.

FINAL ANSWER

$$(a) x = \frac{25}{9}. \quad (b)(i) 4x - 5\sqrt{x} - 44 = 0. \quad (b)(ii) x = 16 \text{ is the only solution.}$$

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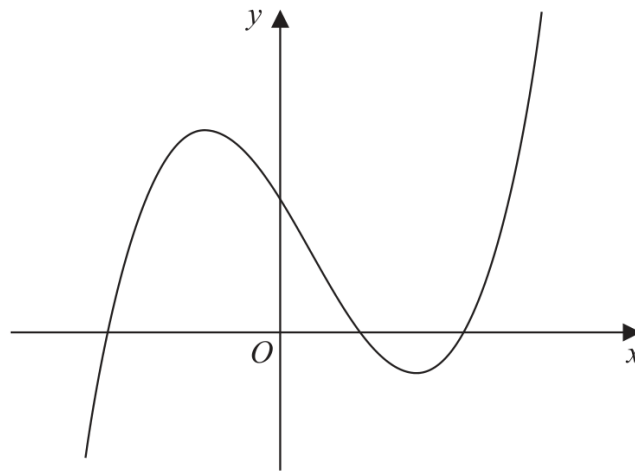


Figure 3

Figure 3 shows a sketch of part of the curve with equation $y = f(x)$, where

$$f(x) = (x + 4)(x - 2)(2x - 9)$$

Given that the curve with equation $y = f(x) - p$ passes through the point with coordinates $(0, 50)$

- (a) find the value of the constant p . (2)

Given that the curve with equation $y = f(x + q)$ passes through the origin,

- (b) write down the possible values of the constant q . (2)

- (c) Find $f'(x)$. (4)

- (d) Hence find the range of values of x for which the gradient of the curve with equation $y = f(x)$ is less than -18 (3)

(Total 11 marks)

Question 22

WMA11/01 January 2022

T12 Differentiation

F03 Equations, inequalities and extrema involving derivatives

V03 Gradient inequality or extremum of the derivative

1

Find p from the translated curve

$f(0) = (4)(-2)(-9) = 72$. Since $y = f(x) - p$ passes through $(0, 50)$, $72 - p = 50$, so $p = 22$.

2

Use roots for $f(x + q)$

If $y = f(x + q)$ passes through the origin, then $f(q) = 0$. From $f(x) = (x + 4)(x - 2)(2x - 9)$, the possible values are $q = -4, 2, \frac{9}{2}$.

3

Expand $f(x)$

$$(x + 4)(x - 2)(2x - 9) = 2x^3 - 5x^2 - 34x + 72.$$

4

Differentiate

$$f'(x) = 6x^2 - 10x - 34.$$

5

Set the gradient less than -18

$$6x^2 - 10x - 34 < -18, \text{ so } 6x^2 - 10x - 16 < 0, \text{ or } 3x^2 - 5x - 8 < 0.$$

6

Solve the inequality

$3x^2 - 5x - 8 = (3x - 8)(x + 1)$. Since the quadratic opens upward, it is negative between the roots: $-1 < x < \frac{8}{3}$.

FINAL ANSWER

$$(a) p = 22. \quad (b) q = -4, 2, \frac{9}{2}. \quad (c) f'(x) = 6x^2 - 10x - 34. \quad (d) -1 < x < \frac{8}{3}.$$

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7. The curve C has equation

$$y = \frac{2}{3}x^3 - 8x^2 + 43x - \frac{20}{3}$$

(a) Show that $\frac{dy}{dx}$ can be written in the form

$$p(x + q)^2 + r$$

where p , q and r are constants to be found.

(5)

(b) Hence state

(i) the minimum value of $\frac{dy}{dx}$

(ii) the value of x at which this minimum value occurs.

(2)

Given that S is the point on C at which the gradient is a minimum,

(c) find the equation of the tangent to C at S , giving your answer in the form $y = mx + c$, where m and c are constants.

(3)

(Total for Question 7 is 10 marks)

Question 23

WMA11/01 January 2026

T12 Differentiation

F03 Equations, inequalities and extrema involving derivatives

V03 Gradient inequality or extremum of the derivative

1

Differentiate

$$y = \frac{2}{3}x^3 - 8x^2 + 43x - \frac{20}{3} \text{ gives } \frac{dy}{dx} = 2x^2 - 16x + 43.$$

2

Complete the square

$$2x^2 - 16x + 43 = 2(x^2 - 8x) + 43 = 2[(x - 4)^2 - 16] + 43.$$

3

Simplify

$$\frac{dy}{dx} = 2(x - 4)^2 + 11.$$

4

Minimum gradient

Since $2(x - 4)^2 \geq 0$, the minimum value of $\frac{dy}{dx}$ is 11.

5

Where it occurs

The minimum occurs when $x - 4 = 0$, so $x = 4$.

6

Find the point S

$$\text{At } x = 4, y = \frac{2}{3}(4)^3 - 8(4)^2 + 43(4) - \frac{20}{3} = 80.$$

7

Tangent equation

The gradient at S is 11, so $y - 80 = 11(x - 4)$.

8

Simplify

$$y = 11x + 36.$$

FINAL ANSWER

$$(a) \frac{dy}{dx} = 2(x - 4)^2 + 11. \quad (b) \min \frac{dy}{dx} = 11 \text{ at } x = 4. \quad (c) y = 11x + 36.$$

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Chord gradients and the derivative limit

WHAT THIS FAMILY ASKS

Calculate a chord gradient using an increment h and show that it approaches the tangent gradient as h tends to zero.

VARIANT ROUTE

V01 Chord gradient tending to the tangent gradient

2 Complete Classified questions

12 total marks

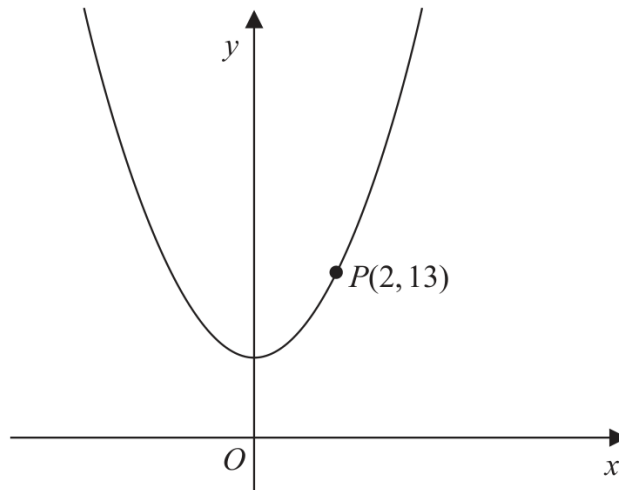


Figure 4

Figure 4 shows part of the curve with equation $y = 2x^2 + 5$

The point $P(2, 13)$ lies on the curve.

- (a) Find the gradient of the tangent to the curve at P . (2)

The point Q with x coordinate $2 + h$ also lies on the curve.

- (b) Find, in terms of h , the gradient of the line PQ . Give your answer in simplest form. (3)
- (c) Explain briefly the relationship between the answer to (b) and the answer to (a). (1)

(Total 6 marks)

Question 24

WMA11/01 October 2019

T12 Differentiation

F04 Chord gradients and the derivative limit

V01 Chord gradient tending to the tangent gradient

1

Differentiate for the tangent gradient

For $y = 2x^2 + 5$, $\frac{dy}{dx} = 4x$. At P , $x = 2$, so the tangent gradient is 8.

2

Find the coordinates of Q

If $x_Q = 2 + h$, then $y_Q = 2(2 + h)^2 + 5 = 13 + 8h + 2h^2$.

3

Find the gradient of PQ

$$\text{gradient } PQ = \frac{(13 + 8h + 2h^2) - 13}{(2 + h) - 2} = \frac{8h + 2h^2}{h} = 8 + 2h.$$

4

Explain the link

As h gets closer to 0, point Q gets closer to P , and the chord gradient $8 + 2h$ tends to the tangent gradient 8.

FINAL ANSWER

(a) 8. (b) $8 + 2h$. (c) As $h \rightarrow 0$, $8 + 2h \rightarrow 8$.

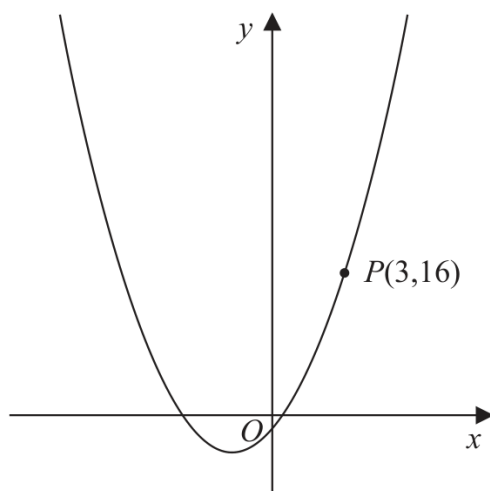


Figure 1

Figure 1 shows part of the curve with equation $y = x^2 + 3x - 2$

The point $P(3, 16)$ lies on the curve.

- (a) Find the gradient of the tangent to the curve at P . (2)

The point Q with x coordinate $3 + h$ also lies on the curve.

- (b) Find, in terms of h , the gradient of the line PQ . Write your answer in simplest form. (3)
- (c) Explain briefly the relationship between the answer to (b) and the answer to (a). (1)

(Total 6 marks)

Question 25

WMA11/01 January 2020

T12 Differentiation

F04 Chord gradients and the derivative limit

V01 Chord gradient tending to the tangent gradient

1

Differentiate for the tangent gradient

For $y = x^2 + 3x - 2$, $\frac{dy}{dx} = 2x + 3$. At $x = 3$, the gradient is $2(3) + 3 = 9$.

2

Find the y-coordinate of Q

For $x_Q = 3 + h$, $y_Q = (3 + h)^2 + 3(3 + h) - 2 = 16 + 9h + h^2$.

3

Calculate the gradient of PQ

gradient $PQ = \frac{(16 + 9h + h^2) - 16}{(3 + h) - 3} = \frac{9h + h^2}{h} = 9 + h$.

4

Explain the relationship

As $h \rightarrow 0$, point Q approaches P, and the chord gradient $9 + h$ approaches the tangent gradient 9.

FINAL ANSWER

(a) 9. (b) $9 + h$. (c) As $h \rightarrow 0$, $9 + h \rightarrow 9$.

Global curve synthesis with tangents

WHAT THIS FAMILY ASKS

Combine factorised or expanded curve structure with a tangent equation to locate another intersection or interpret the whole curve.

VARIANT ROUTE

V01 Tangent re-intersects the original curve

1 Complete Classified questions

10 total marks

10.

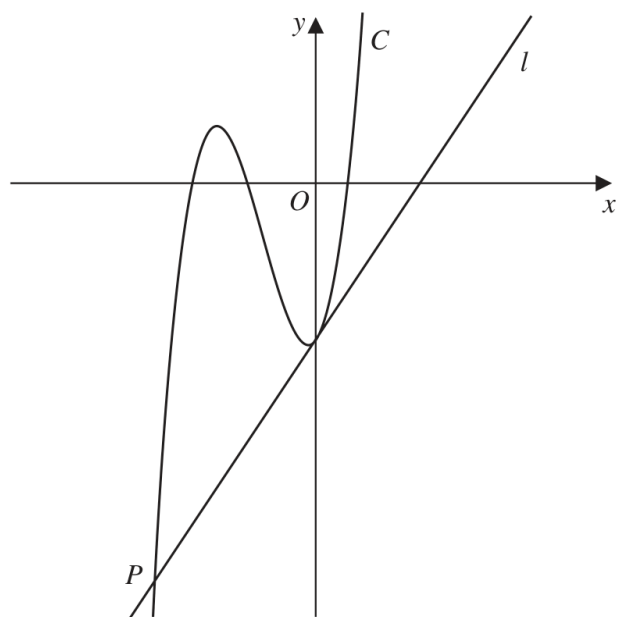


Figure 4

Figure 4 shows a sketch of part of the curve C with equation $y = f(x)$, where

$$f(x) = (3x + 20)(x + 6)(2x - 3)$$

- (a) Use the given information to state the values of x for which

$$f(x) > 0$$

(2)

- (b) Expand $(3x + 20)(x + 6)(2x - 3)$, writing your answer as a polynomial in simplest form.

(3)

The straight line l is the tangent to C at the point where C cuts the y -axis.

Given that l cuts C at the point P , as shown in Figure 4,

- (c) find, using algebra, the x coordinate of P

(Solutions based on calculator technology are not acceptable.)

(5)

(Total for Question 10 is 10 marks)

Question 26

WMA11/01 January 2023

T12 Differentiation

F05 Global curve synthesis
with tangentsV01 Tangent re-intersects the
original curve

1

Use the roots

$f(x) = (3x + 20)(x + 6)(2x - 3)$ has roots $x = -\frac{20}{3}$, $x = -6$, and $x = \frac{3}{2}$.

2

State where $f(x)$ is positive

Since the cubic has positive leading coefficient, $f(x) > 0$ for $-\frac{20}{3} < x < -6$ and for $x > \frac{3}{2}$.

3

Expand the first two brackets

$$(3x + 20)(x + 6) = 3x^2 + 38x + 120.$$

4

Complete the expansion

$$(3x^2 + 38x + 120)(2x - 3) = 6x^3 + 67x^2 + 126x - 360.$$

5

Find the tangent at the y-axis

$f'(x) = 18x^2 + 134x + 126$, so at $x = 0$ the tangent gradient is 126. Also $f(0) = -360$.

6

Write the tangent equation

The tangent line is $y = 126x - 360$.

7

Find the other intersection

Set $6x^3 + 67x^2 + 126x - 360 = 126x - 360$. This gives $6x^3 + 67x^2 = 0$, so $x^2(6x + 67) = 0$.

8

Choose point P

$x = 0$ is the tangent point on the y-axis, so the other intersection P has $x = -\frac{67}{6}$.

FINAL ANSWER

(a) $-\frac{20}{3} < x < -6$ **or** $x > \frac{3}{2}$. (b) $6x^3 + 67x^2 + 126x - 360$. (c) $x_P = -\frac{67}{6}$.

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